Chapter 3
Logarithms in Earth and Environmental Sciences

3.1 Earthquake!

Problem 3.7 *The Richter Scale is one way to quantify the magnitude of Earthquakes. The magnitude of an Earthquake is independent of distance from the Earthquake as it refers to the magnitude of the Earthquake at the epicentre.*

$$m_l = \log_{10}\left(\frac{A}{A_0}\right)$$

where A_0 is the amplitude of the a wave at the epicentre that would just be detected (mm) and A is the amplitude of the wave measured (mm).
A_0 is given by

$$-\log_{10}(A_0) \cong 1.4 + 0.16\sqrt{D}$$

where D is the distance to the epicentre in km.
In this exercise, you will convert the amplitudes produced by an Earthquake at distance D to an equivalent magnitude for if the Earthquake occurred locally (where $D = 0$).

The 1886 Earthquake in Charlston, South Carolina is estimated to be a 7.3 on the Richer magnitude scale. Complete the analysis below.

1. *Calculate the amplitude, A, measured $D = 100$ km away from the epicentre.*
2. *If this amplitude was measured at the epicentre, $D = 0$ km, what would the magnitude be?*
3. *Calculate the amplitude measured $D = 1000$ km away in New York City.*
4. *If this amplitude in New York City was measured at the epicentre of an Earthquake what would the magnitude be ?*

3.1.1 Notes

1. Rearrange the equation for A and calculate A using available data.
   - Substitute the equation for $-\log_1 0\,(A_0)$ into the equation for m_l

$$m_l = \log_{10}(A) + 1.4 + 0.16\sqrt{D}$$

- Rearrange for A. Start by subtracting 1.4 and $0.16\sqrt{D}$ from both sides:

$$\begin{aligned} m_l - 1.4 - 0.16\sqrt{D} &= \log_{10}(A) \\ &\therefore \\ 10^{m_l - 1.4 - 0.16\sqrt{D}} &= A \end{aligned}$$

- With $D = 100$ km this results in $A \cong 20,000$ mm

2. Using $A = 20,000$ mm above, and the value of A_0 at $D = 0$ calculate the resulting magnitude.

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$$\begin{aligned} m_l &= \log_{10}(20000) + 1.4 \\ m_l &= 5.7 \end{aligned}$$

3. Same as for question 1, but use $D = 1000$ km

- Substitute the equation for $-\log_1 0\,(A_0)$ into the equation for m_l

$$m_l = \log_{10}(A) + 1.4 + 0.16\sqrt{D}$$

- Rearrange for A. Start by subtracting 1.4 and $0.16\sqrt{D}$ from both sides:

$$\begin{aligned} m_l - 1.4 - 0.16\sqrt{D} &= \log_{10}(A) \\ &\therefore \\ 10^{m_l - 1.4 - 0.16\sqrt{D}} &= A \end{aligned}$$

- With $D = 1000$ km this results in $A \cong 6.9$ mm

4. Using $A = 6.9$mm above, and the value of A_0 at $D = 0$ calculate the resulting magnitude.

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$$\begin{aligned} m_l &= \log_{10}(6.9) + 1.4 \\ m_l &= 2.24 \end{aligned}$$

3.2 Problems

Atmospheric Science 1: The freezing of cloud droplets in the atmosphere is a stochastic process and obeys a law that is similar to radioactive decay.

$$N = N_0 \exp(-Jt)$$

where N is the number of unfrozen droplets at time t; and N_0 is the initial number of unfrozen droplets.
At a temperature below zero degrees celcius take $J = 0.1\ \text{s}^{-1}$. How long would it take for the unfrozen droplets to decrease to 1% of the initial value?

Atmospheric Science 2: In the atmosphere the pressure varies with altitude approximately according to the hydrostatic equation.

$$P = P_0 \exp\left(-\frac{z}{H}\right)$$

where P is the pressure at height z, P_0 is the initial pressure, z is the height in metres and $H = 8000$ m is the 'scale height' of the atmosphere.
Additionally, under certain conditions, the quantity θ, given below, does not change

$$\theta = T\left(\frac{10^5}{P}\right)^{0.286}$$

where T is the temperature and P is the pressure.

If a parcel of air has a temperature $T = 285$ K and a pressure $P = 9 \times 10^4$ Pa, calculate the quantity θ. A cloud may form if the temperature drops by 5 K.

Calculate the resulting pressure drop, if θ stays constant during this process.

How far must the parcel have risen in metres for the cloud to form? (hint use the hydrostatic equation above).

Population growth: In 2019, according to `https://www.worldometers.info/world-population/`, the World's population is growing at 1.08% per year (i.e. multiplying by factor of 1.0108 each year.

How many years will it take for the population double?

3.2.1 Notes on Problems

Atmospheric Science 1: Rearrange the nucleation equation for t (time) and put values in

$$\begin{aligned} \frac{N}{N_0} &= \exp(-Jt) \\ \ln\left(\frac{N}{N_0}\right) &= (-Jt) \\ \therefore & \\ t &= -\frac{1}{J}\ln\left(\frac{N}{N_0}\right) \\ t &= -\frac{1}{0.1}\ln(0.01) \end{aligned}$$

which results in $t = 46$ s

Atmospheric Science 2: Firstly, calculate θ from the initial temperature and pressure.

$$\begin{aligned} \theta &= 285\left(\frac{10^5}{9\times 10^4}\right)^{0.286} \\ \therefore & \\ \theta &= 293.72K \end{aligned}$$

Now rearrange the θ equation for pressure:

$$\begin{aligned} P &= 10^5\left(\frac{T}{\theta}\right)^{1/0.286} \\ \therefore & \\ P &= 10^5\left(\frac{280}{293.72}\right)^{1/0.286} \\ P &= 8.46\times 10^4 Pa \end{aligned}$$

Now, use the hydrostatic equation to solve for z and use the two pressures.

$$\begin{aligned} P &= P_0\exp\left(-\frac{z}{H}\right) \\ \frac{P}{P_0} &= \exp\left(-\frac{z}{H}\right) \\ \ln\left(\frac{P}{P_0}\right) &= -\frac{z}{H} \\ \therefore & \\ z &= -H\ln\left(\frac{P}{P_0}\right) \\ \therefore & \\ z = &= -8000\ln\left(\frac{8.46\times 10^4}{9\times 10^4}\right) \\ \therefore & \\ P &= = 495m \end{aligned}$$

Population growth: For this one you will have to write down your own equation. The population n-years into the future, P, is the current population, P_0, multiplied by 1.0108^n

$$P = P_0 \times (1.0108)^n$$

If the population doubles then $P = 2P_0$. Put this in and solve for n

$$\begin{aligned} 2\cancel{P_0} &= \cancel{P_0} \times (1.0108)^n \\ \ln 2 &= \ln (1.0108)^n \end{aligned}$$

Using the 3rd law of logarithms

$$\begin{aligned} \ln 2 &= n \ln (1.0108) \\ \therefore \\ n &= \frac{\ln 2}{\ln (1.0108)} \\ n &= 65 years \end{aligned}$$

3.2.2 Homework and reading for next time

Background reading:

- Croft and Davison (2006, chapters on 'The exponential function', 'The logarithm function') OR 2.16,2.17,2.18,3.4,3.5 and 3.6 of the *Foundation Maths Support Pack*.

Do this weeks assessment on Blackboard:

- 'The exponential and the logarithm functions'.
- 'Logarithms in Earth and Environmental Science'.